UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 015 · Applications of LDEs: simple harmonic oscillation · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S015·11	Unit 2 deck 015 ("Applications of DE – SHM"), slide 11 (rendered .pptx)
T015 00:11:48	Unit 2 AV-summary transcript "Application problems on linear differential equations – Simple harmonic oscillation" (16.3 min), segment start
CW U2 P37	Unit 2 class-work problem 37

Contents

1. 1 · The spring–mass model and simple harmonic motion
2. 2 · Solving spring problems
3. 3 · The damped oscillator
4. Past-paper questions

1 The spring–mass model and simple harmonic motion

Definitions

Equilibrium position. The position where the downward pull of gravity equals the upward restoring force of the spring: $mg = ks$. S015·5 · T015 00:02:29

Hooke's law. The restoring force is proportional to the displacement and acts in the opposite direction: $-k(s + x)$. S015·6 · T015 00:03:37

Spring (force) constant k. The proportionality constant in Hooke's law; force per unit stretch (lb/ft or N/m). S015·6 · T015 00:03:55

Simple harmonic motion. The motion $x(t) = c_1 \cos \omega t + c_2 \sin \omega t$ of the mass about its equilibrium position. S015·8 · T015 00:06:46

Angular frequency ω. $\omega = \sqrt{k/m}$, from writing the equation as $\frac{d^2x}{dt^2} + \omega^2 x = 0$. S015·7 · T015 00:05:33

Amplitude A. "A is called amplitude of the motion and gives the maximum displacement of the mass from its equilibrium position." S015·9 · T015 00:07:22

Period T. The time taken for one complete oscillation: $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$. S015·9 · T015 00:07:40

Natural frequency f. The reciprocal of the period: $f = \frac{1}{T} = \frac{\omega}{2\pi}$ oscillations per second. S015·9 · T015 00:07:58

Statements in the class material

> *"From Hooke's Law, the restoring force is given by $-k(s+x)$ (the restoring force is proportional to displacement and acts in the opposite direction of displacement). By Newton's Second Law, the forces acting on the system $= ma$, i.e, $mg + (-k(s+x)) = ma$."* S015·6

> *"But at equilibrium, $mg = ks$. Therefore, $ma = -kx$. ... $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$. This equation can also be written as $\frac{d^2x}{dt^2} + \omega^2 x = 0$ where $\omega = \sqrt{\frac{k}{m}}$ called angular frequency."* S015·7

> *"The general solution of the differential equation is given by $x(t) = c_1\cos\omega t + c_2\sin\omega t$. $x(t)$ gives the position of the mass at any instant. The motion of the mass is Simple Harmonic Motion."* S015·8

Key points

1. Two forces act on the mass: gravity mg (down) and the restoring force $-k(s+x)$; at equilibrium $mg = ks$, so only $-kx$ is left. ESA ★★ S015·6 · S015·7 · T015 00:03:37 · T015 00:05:03
2. x is measured from the equilibrium position, not from the natural length; that is why mg disappears from the equation. ESA ★★ S015·5 · T015 00:03:13
3. The equation is homogeneous with constant coefficients; its roots $\pm i\omega$ are purely imaginary, so the solution is the CF alone (no PI). ESA ★★ T015 00:05:53 · T015 00:06:10 · T015 00:06:28
4. "You should remember all these formulae": $\omega = \sqrt{k/m}$, $A = \sqrt{c_1^2 + c_2^2}$, $T = 2\pi\sqrt{m/k}$, $f = \frac{\omega}{2\pi}$. ESA ★★ S015·9 · T015 00:07:58
5. The period does not depend on the amplitude or on how the motion starts, only on m and k. ESA ★ (added for clarity) S015·9

Why?

Why does gravity not appear in $\ddot{x} + \frac{k}{m}x = 0$?
Because x is measured from equilibrium, where $mg = ks$ already. In $mg - k(s+x)$ the two constant forces cancel and only $-kx$ is left.
S015·7 · T015 00:05:03

Why are the motions sinusoidal?
The auxiliary equation $m^2 + \omega^2 = 0$ has roots $\pm i\omega$ with zero real part, so the CF is $c_1\cos\omega t + c_2\sin\omega t$: no growth and no decay.
T015 00:06:28 · S015·8

Exam traps

Exam trap. Amplitude is $\sqrt{c_1^2 + c_2^2}$ (a sum of squares), not "c_1 square by c_2 square" as said at T015 00:07:22. ADDED FOR CLARITY S015·8 · T015 00:07:22

Exam trap. At equilibrium $mg = ks$ with the static stretch s, not kx (T015 00:02:46): x is the displacement from equilibrium. ADDED FOR CLARITY S015·5 · T015 00:02:46

2 Solving spring problems

Definitions

Free motion. Motion with no external force and no damping: $\ddot{x} + \frac{k}{m}x = 0$. S015·10 · T015 00:08:48

Key points

1. Use the static data first: the weight gives $m = W/g$, and the stretch it causes gives $k = W/s$. The later data (pulled down, velocity) fix c_1, c_2. ESA ★★ S015·11 · T015 00:09:25
2. Convert inches to feet (12 in = 1 ft) before using $g = 32$ ft/s²: 6 in = 0.5 ft, 4 in = $\frac{1}{3}$ ft, 5 in = $\frac{5}{12}$ ft. ESA ★★ S015·11 · S015·13 · S015·20 · T015 00:10:47
3. Downward is positive. An upward velocity of 2 ft/s is $\dot{x}(0) = -2$; a mass raised 5 in has $x(0) = -\frac{5}{12}$ ft. ESA ★★ S015·14 · S015·20 · T015 00:14:07 · T015 00:14:24
4. Differentiate $x(t)$ before using a velocity condition: $\dot{x} = -c_1\omega \sin \omega t + c_2\omega \cos \omega t$, so $\dot{x}(0) = c_2\omega$. ESA ★★ S015·13 · T015 00:13:31
5. Released from rest means $\dot{x}(0) = 0$, so $c_2 = 0$ and $x = x(0)\cos \omega t$; the amplitude is then $|x(0)|$. ESA ★★ S015·20 · S015·21 · CW U2 P37
6. The maximum velocity $A\omega$ is reached when the mass passes through equilibrium. ESA ★★ ADDED FOR CLARITY S015·22 · CW U2 P37

Why?

Why is an upward velocity negative?
The deck takes downward displacement as positive; a velocity in the opposite (upward) direction therefore carries a minus sign.
T015 00:14:07 · T015 00:14:24 · S015·20

Exam traps

Exam trap. Mixing inches and feet: with $g = 32$ ft/s², every length must be in feet ($s = 6$ in = 0.5 ft gives $k = 24$; using 6 gives $k = 2$, wrong). ADDED FOR CLARITY S015·11

Exam trap. $c_2 = \frac{\dot{x}(0)}{\omega}$, not $\dot{x}(0)$: in the 12 lb example $8c_2 = -2$ gives $c_2 = -\frac{1}{4}$ (the lecture first says "minus 4", T015 00:14:59). ADDED FOR CLARITY S015·14 · T015 00:14:59

Exam trap. Weight is not mass: a 12 lb weight has $m = \frac{12}{32}$ slug. In SI, a 20 kg weight is a force of $20g = 196$ N. ADDED FOR CLARITY S015·11 · CW U2 P37

Exam trap. A body raised above equilibrium starts at a negative x ($-\frac{5}{12}$ ft on slide 015·20); forgetting the sign flips the phase. ADDED FOR CLARITY S015·20

3 The damped oscillator

Definitions

Damping (resistive) force. A force opposing the motion and proportional to the velocity; on slide 015·15 its magnitude is $2\lambda m n \dot{x}$. S015·15

Key points

1. A resistance proportional to velocity adds the term $2\lambda n\dot{x}$; for $0 < \lambda < 1$ the mass still oscillates, but inside the envelope $\pm Ce^{-\lambda n t}$. ESA ★ S015·15 · S015·16
2. Damping lowers the frequency: $\omega_d = n\sqrt{1-\lambda^2} < n$. ESA ★ added for clarity S015·16
3. With $\lambda = 1$ (critical) or $\lambda > 1$ (overdamped) the roots are real and the mass returns to equilibrium without oscillating. ESA ★ added for clarity S015·16

Why?

Why is the damped motion called a twin of the LCR circuit?
$m\ddot{x} + c\dot{x} + kx = F(t)$ and $L\ddot{Q} + R\dot{Q} + \frac{Q}{C} = E(t)$ are the same equation with $m \leftrightarrow L$, $c \leftrightarrow R$, $k \leftrightarrow \frac{1}{C}$ (table C1.2).
added for clarity S015·4 · T015 00:00:32

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). Simple harmonic oscillation (the spring–mass problem) was never asked in any of them: 0 of 8 papers. It is in the syllabus ("Application problems – ... Oscillations of a spring") and the class-work sheet has one problem (CW P37), so the method is ★★ and the formulas $\omega = \sqrt{k/m}$, $T = 2\pi/\omega$, $f = 1/T$ are ★★ because the lecturer stresses them ("you should remember all these formulae", T015 00:07:58). The derivation and the damped oscillator are ★. The applications question the papers do ask is the LCR circuit (class 014, Jan 2024 sets 3 and 4).

No past-paper questions in scope.